SOC LAB PROJECT REPORT

Detection of Brute-Force Attack and File Integrity Monitoring using Wazuh SIEM

Kali Linux · Windows 10 · Ubuntu Wazuh Server

Prepared By	Rakesh A R
Role	Cybersecurity Enthusiast
Email	rakesharofficial06@gmail.com
LinkedIn	linkedin.com/in/rakesh-a-r-595517288
GitHub	github.com/zeroday-studio

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SIEM Platform	Wazuh SIEM v4.x
Lab Type	Self-Learning Home Lab
Report Classification	Portfolio / Resume Project

1. Objective

To simulate a brute-force attack using Kali Linux and monitor file integrity changes on a Windows 10 system using Wazuh SIEM. The lab demonstrates real-world SOC detection capabilities including authentication failure analysis and real-time file activity monitoring — built entirely as a self-learning home lab project.

2. Lab Setup

2.1 Network Architecture

Machine	Role	IP Address	Notes
Kali Linux	Attacker	192.168.10.5	Hydra brute-force tool
Windows 10	Victim / Agent	192.168.10.10	Wazuh agent installed
Ubuntu	Wazuh SIEM Server	192.168.10.20	Log collector & dashboard

2.2 Tools Used

- Kali Linux — Attacker machine
- Hydra v9.6 — Dictionary brute-force tool targeting SMB (port 445)
- Wazuh SIEM — Security event collection, correlation, and alerting
- Windows Event Logs — Authentication failure events (Event ID 4625)
- File Integrity Monitoring (FIM) — Wazuh module for file change detection

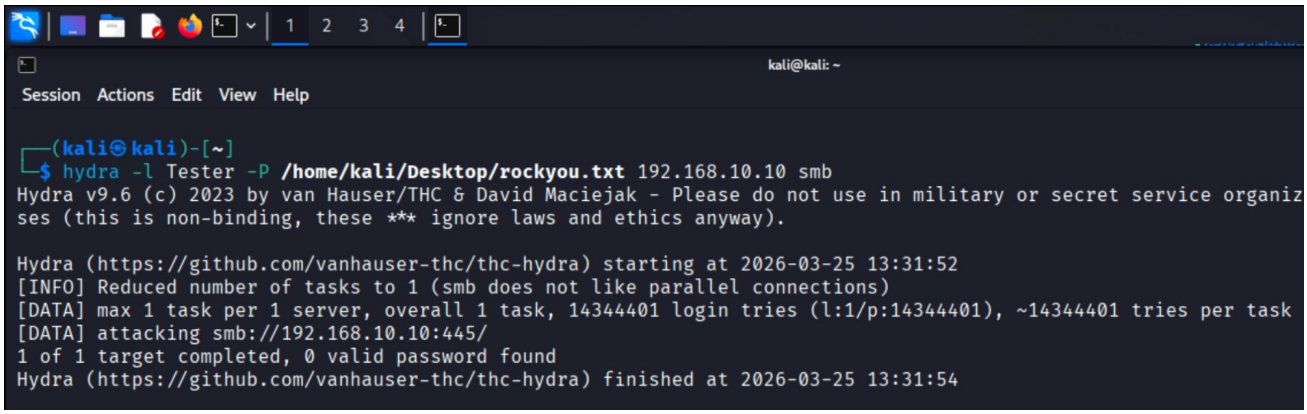
3. Brute-Force Attack Simulation

3.1 Attack Description

A dictionary-based brute-force attack was launched from the Kali Linux machine (192.168.10.5) targeting the SMB service (port 445) on the Windows 10 victim (192.168.10.10). Hydra v9.6 was used with the rockyou.txt wordlist to attempt multiple password combinations for the user account 'Tester'.

3.2 Hydra Command Executed

```
hydra -l Tester -P /home/kali/Desktop/rockyou.txt 192.168.10.10 smb
```



Screenshot 1: Hydra v9.6 brute-force attack running on Kali Linux targeting SMB (192.168.10.10:445)

4. Log Collection

The Wazuh agent installed on the Windows 10 machine continuously collected Windows Security Event Logs and forwarded them to the Wazuh SIEM server at 192.168.10.20. The agent monitored the Windows Security channel and Wazuh automatically applied detection rules to identify suspicious authentication patterns in real time.

5. Detection Method

5.1 Windows Event IDs

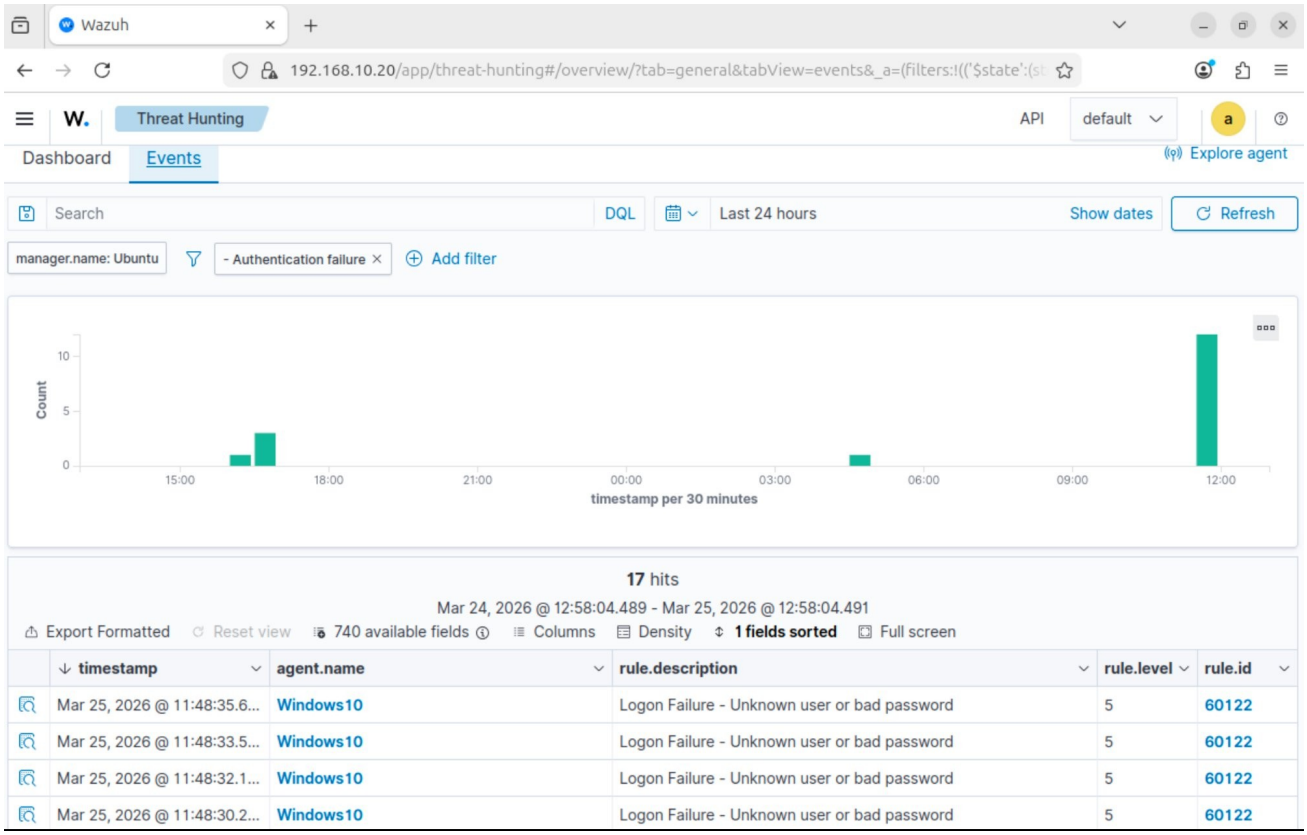
Event ID	Event Name	Significance
4625	Failed Logon	Authentication failure — wrong password
4624	Successful Logon	Successful login after brute-force
4776	Credential Validation	NTLM credential validation attempt

5.2 Logon Type Analysis

Logon Type	Name	Meaning
Type 3	Network	Remote login over network (SMB) — seen in this lab
Type 2	Interactive	Local keyboard login
Type 10	RemoteInteractive	RDP / Terminal Services login

6. Failed Login Detection

Multiple failed login attempts (Event ID 4625, Rule ID 60122 — Logon Failure) were detected in the Wazuh Threat Hunting dashboard. A total of 17 hits were recorded with a clear spike at 12:00 confirming the brute-force attack burst from the Windows10 agent.



Screenshot 2: Wazuh Threat Hunting — 17 failed authentication events (Rule 60122) with spike at 12:00

7. Detailed Log Analysis

7.1 Event Fields — Part 1 (Agent & Authentication Data)

Expanding a failed login event in the Wazuh Discover view reveals full event metadata confirming the attacker source IP, target username, and logon type:

Discover wazuh-alerts-4.x-2026.03.25#eLAwJZ0BZJ5SBjJ-AeLN

Table JSON

@timestamp	Mar 25, 2026 @ 13:29:54.853
_index	wazuh-alerts-4.x-2026.03.25
agent.id	001
agent.ip	192.168.10.10
agent.name	Windows10
data.win.eventdata.authenticationPackageName	NTLM
data.win.eventdata.failureReason	%%2313
data.win.eventdata.ipAddress	192.168.10.5
data.win.eventdata.ipPort	59658
data.win.eventdata.keyLength	0
data.win.eventdata.logonProcessName	NtLmSsp
data.win.eventdata.logonType	3
data.win.eventdata.processId	0x0
data.win.eventdata.status	0xc000006d
data.win.eventdata.subStatus	0xc000006a
data.win.eventdata.subjectLogonId	0x0
data.win.eventdata.subjectUserSid	S-1-0-0
data.win.eventdata.targetUserName	Tester
data.win.eventdata.targetUserSid	S-1-0-0

Screenshot 3a: Wazuh event detail — Source IP 192.168.10.5, Username: Tester, Logon Type: 3, Auth Package: NTLM

7.2 Event Fields — Part 2 (System & Security Data)

Discover wazuh-alerts-4.x-2026.03.25#eLAwJZ0BZJ5SBjJ-AeLN

data.win.eventdata.workstationName	\\\\192.168.10.5
data.win.system.channel	Security
data.win.system.computer	Windows
data.win.system.eventID	4625
data.win.system.eventRecordID	16317
data.win.system.keywords	0x8010000000000000
data.win.system.level	0
data.win.system.message	"An account failed to log on. Subject: Security ID: S-1-0-0 Account Name: - Account Domain: - Logon ID: 0x0
data.win.system.opcode	0
data.win.system.processID	568
data.win.system.providerGuid	{54849625-5478-4994-a5ba-3e3b0328c30d}
data.win.system.providerName	Microsoft-Windows-Security-Auditing
data.win.system.severityValue	AUDIT_FAILURE
data.win.system.systemTime	2026-03-26T01:53:00.6976068Z
data.win.system.task	12544
data.win.system.threadID	884
data.win.system.version	0

Screenshot 3b: Wazuh event detail — Event ID 4625, Severity: AUDIT_FAILURE, Message: An account failed to log on

7.3 Key Fields Summary

Field	Value
Event ID	4625
Logon Type	3 (Network — remote SMB attack)
Target Username	Tester
Source IP	192.168.10.5 (Kali Linux)
Auth Package	NTLM
Severity	AUDIT_FAILURE
Wazuh Rule ID	60122 — Logon Failure
Agent	Windows10 (192.168.10.10)

8. Brute-Force Detection Summary

The Wazuh dashboard confirmed the brute-force attack pattern by correlating 17 failed authentication events from the same source IP within a short window. The events graph clearly shows the burst of attack attempts.

Metric	Observation
Total Failed Attempts	17 hits detected by Wazuh
Source IP	192.168.10.5 (Kali Linux)
Target Username	Tester
Wazuh Rule	60122 — Logon Failure
MITRE Technique	T1110 — Brute Force
Attack Confirmed	Yes — spike visible at 12:00 in events graph

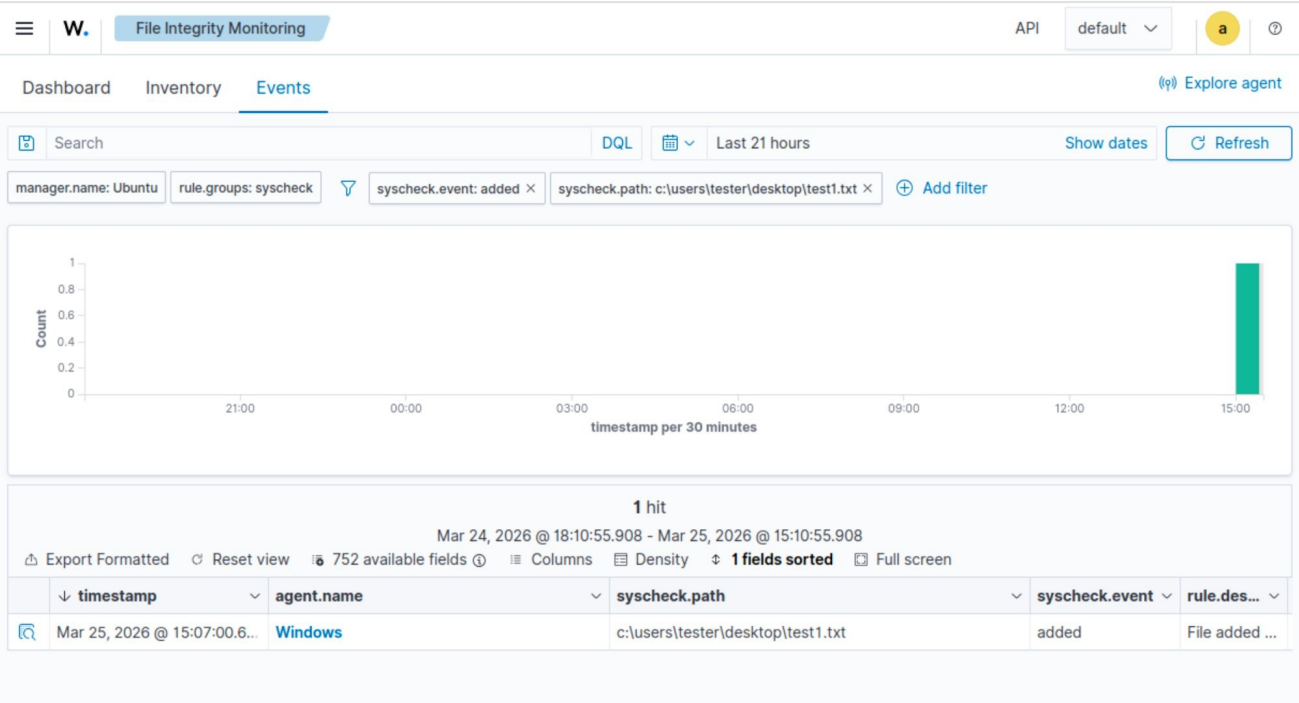
9. File Integrity Monitoring (FIM)

9.1 Overview

The Wazuh FIM module was configured on the Windows 10 agent to monitor the Desktop directory (C:\Users\Tester\Desktop) for any file activity. FIM uses checksums and metadata to detect file additions, modifications, and deletions in real time.

9.2 Test 1 — File Creation (syscheck.event = added)

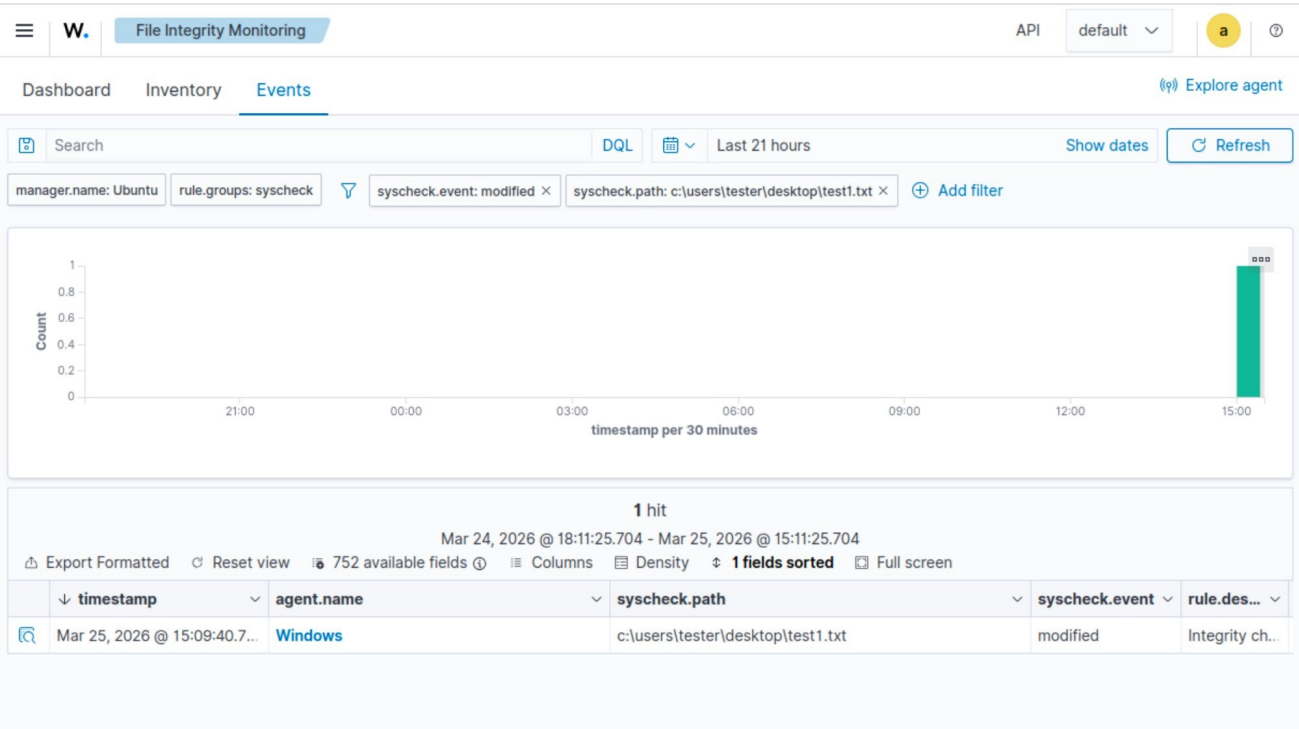
A new file named test1.txt was created on the Windows 10 Desktop. Wazuh FIM immediately detected the addition and generated a 'File added' alert.



Screenshot 5: Wazuh FIM — syscheck.event = added (test1.txt created on Desktop, 1 hit detected)

9.3 Test 2 — File Modification (syscheck.event = modified)

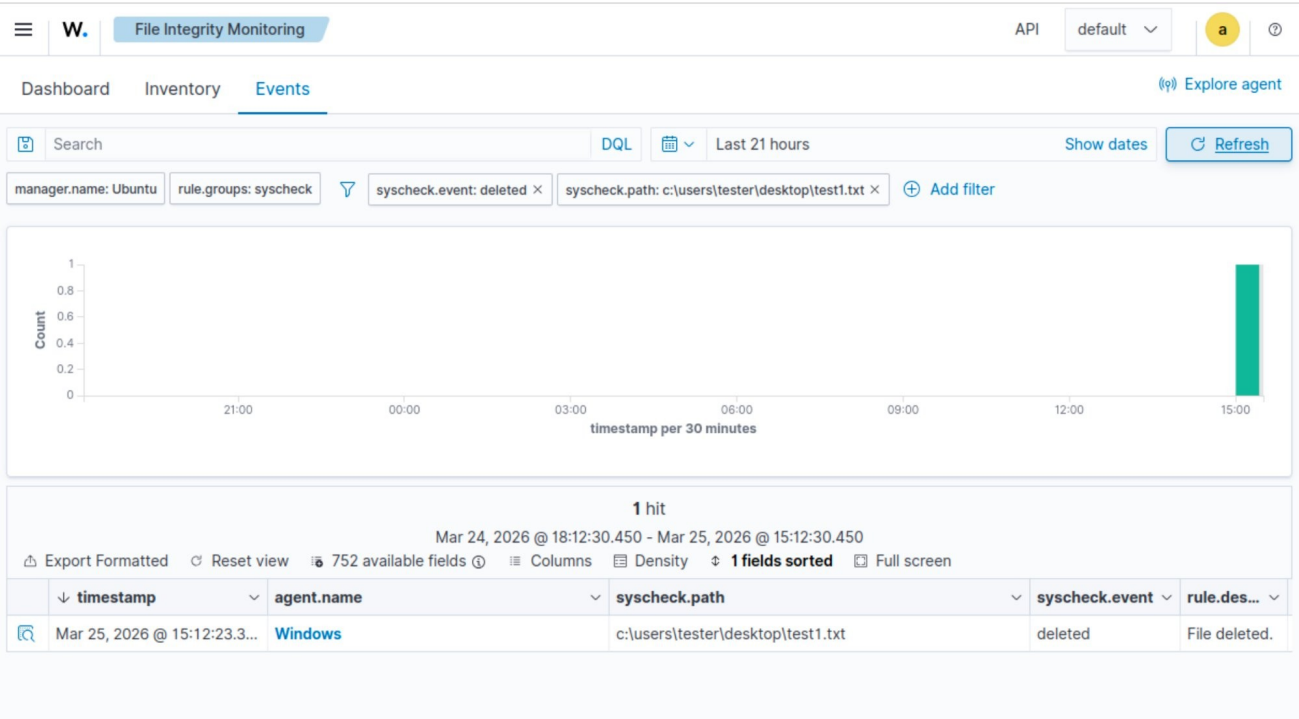
The contents of test1.txt were modified. Wazuh FIM detected the hash mismatch and raised an 'Integrity checksum changed' alert.



Screenshot 6: Wazuh FIM — syscheck.event = modified (test1.txt content changed, integrity checksum mismatch)

9.4 Test 3 — File Deletion (syscheck.event = deleted)

The file test1.txt was deleted using Command Prompt. Wazuh FIM detected the removal and generated a 'File deleted' alert.



Screenshot 7: Wazuh FIM — syscheck.event = deleted (test1.txt removed, rule.des = File deleted)

9.5 FIM Results Summary

Test	syscheck.event	File	Wazuh Rule	Result
File Creation	added	test1.txt	550 — File added	✓
File Modification	modified	test1.txt	550 — Integrity checksum	✓
File Deletion	deleted	test1.txt	553 — File deleted	✓

10. Analysis

10.1 Brute-Force Attack Analysis

1. Each failed SMB login generated a Windows Event ID 4625 with Logon Type 3, confirming a remote network-based attack
2. Wazuh correlated 17 failures from source IP 192.168.10.5 within a short window, matching brute-force rule 60122

3. NTLM authentication package and NtLmSsp logon process confirm SMB-based credential attack
4. The attacker IP, targeted username (Tester), and timestamps were all captured in the alert details

10.2 FIM Analysis

5. File creation detected immediately — syscheck.event = added at 15:07:00
6. File modification detected through hash mismatch — syscheck.event = modified at 15:09:40
7. File deletion detected and flagged — syscheck.event = deleted at 15:12:23
8. All three events confirmed from the Windows agent with complete audit trail

11. MITRE ATT&CK Mapping

Technique ID	Technique Name	Tactic	Detection
T1110	Brute Force	Credential Access	Wazuh Rule 60122
T1078	Valid Accounts	Defense Evasion	Event ID 4625 / 4624
T1565.001	Stored Data Manipulation	Impact	Wazuh FIM Rule 550/553
T1083	File & Dir Discovery	Discovery	FIM syscheck monitoring

12. Recommendations

Based on the findings from this lab, the following security recommendations are proposed to defend against the detected attack techniques:

Finding	Risk	Recommendation	Priority
Brute-force via SMB	Account takeover	Block SMB (port 445) externally; restrict to internal network only	**High**
No account lockout policy	Unlimited login attempts	Enable Account Lockout Policy — lock after 5 failed attempts for 15 minutes	**High**
Weak/guessable passwords	Easy credential cracking	Enforce strong password policy — minimum 12 characters, complexity required	**High**
No Multi-Factor Auth (MFA)	Single point of failure	Implement MFA for all user accounts, especially admin accounts	**High**
No IP-based blocking	Persistent attack source	Deploy fail2ban or Wazuh Active Response to auto-block attacker IPs after 5 fails	**Medium**
FIM on Desktop only	Limited file coverage	Extend FIM monitoring to C:\	**Medium**

		Windows\System32, C:\Program Files, and registry	
No alerting/notification	Delayed incident response	Configure Wazuh email or Slack alerts for Level 7+ events for real-time response	**Medium**
No network segmentation	Lateral movement risk	Segment attacker and victim VMs — use firewall rules to limit cross-machine access	**Low**

12.1 Quick Wins (Implement First)

- Enable Account Lockout Policy on Windows — prevents unlimited brute-force attempts
- Block port 445 (SMB) from external/untrusted networks at the firewall level
- Enable Wazuh Active Response to automatically block IPs with repeated failed logins
- Set up email alerts in Wazuh for any alert Level 7 and above

13. Key Learnings

Brute-Force Detection	File Integrity Monitoring
• Event ID 4625 = Failed login attempt • Logon Type 3 = Remote network attack • Repeated failures from same IP = Brute force • Source IP captured for investigation • MITRE T1110 mapped to the attack	• FIM detects add, modify, delete events • Hash comparison detects even minor changes • Rule 550 = file changed, Rule 553 = deleted • syscheck.event shows exact change type • Real-time monitoring provides instant alerts

14. Conclusion

This self-learning SOC lab project successfully demonstrated two key cybersecurity monitoring capabilities using the Wazuh SIEM platform:

9. Brute-Force Attack Detection — Wazuh effectively detected 17 repeated failed login attempts (Event ID 4625, Logon Type 3) from the Kali Linux attacker (192.168.10.5) and generated alerts identifying the attacker IP, target username, and attack pattern.
10. File Integrity Monitoring — Wazuh FIM successfully detected all three types of file activity (creation, modification, and deletion) of test1.txt on the Windows Desktop, providing a complete real-time audit trail.

This project was built entirely as a self-learning initiative — demonstrating hands-on cybersecurity skills, SOC analyst workflows, and the ability to independently deploy, configure, and operate a real-world SIEM platform without institutional support.

Rakesh A R

Cybersecurity Enthusiast | SOC & SIEM Practitioner

✉️ rakesharofficial06@gmail.com 🔗 LinkedIn 💻 GitHub: zeroday-studio

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